

Rural Fire Dataset: An Explainable Artificial Intelligence (XAI) Study

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Abstract—This work applies Explainable Artificial Intelligence (XAI) techniques to the Portuguese Rural Fire Dataset (PRFD) to analyse the prediction of intentional versus non-intentional fire events. A supervised classification pipeline is implemented using both interpretable and black-box models, complemented by post-hoc explanation methods. The study examines the contribution of different XAI classes—simplification-based, feature-based, and example-based—and evaluates the quality of the resulting explanations through functionally grounded metrics, demonstrating that while feature-based approaches offer high fidelity, example-based methods provide superior user interpretability. The findings also highlight the role of XAI in improving transparency and interpretability in fire-related predictive modelling.

Index Terms—Explainable Artificial Intelligence, Rural Fire Dataset, Intentionality Prediction, SHAP, Counterfactual Explanations, XAI metrics.

INTRODUCTION

Wildfires constitute a persistent environmental challenge in mainland Portugal, with thousands of events recorded every year by national authorities. Understanding the conditions associated with fire ignition is essential for prevention, monitoring, and operational planning. In this work we use a large-scale national rural fire database covering mainland Portugal over several decades, which provides detailed information on the timing, location and characteristics of each recorded event.

We formulate the problem as a supervised binary classification task: predicting whether a rural fire was recorded as *intentional* or *non-intentional*. Beyond predictive accuracy, the goal is to analyse which factors most strongly contribute to this classification. To achieve this, we integrate explainable artificial intelligence (XAI) techniques throughout the modelling pipeline, allowing us to interpret the behaviour of both interpretable and black-box models.

The project is organised into four tasks. First, pre-modelling XAI techniques are applied to characterise the dataset and identify relevant patterns. Second, glass-box models are trained to establish interpretable baselines. Third, a black-box model is developed, and post-modelling XAI methods (simplification-based, feature-based, and example-based) are used to generate local and global explanations. Finally, the quality of these explanations is assessed using functionally grounded metrics that quantify fidelity and robustness.

TASK 1 – PRE-MODELLING XAI

1.1 Dataset description and pre-processing

The analysis relies on the Portuguese Rural Fire Database (PRFD) for mainland Portugal, compiled by the Portuguese Forest Service and described in detail by Pereira et al. [1]. The PRFD covers the period 1980–2005 and contains more than 450 000 rural fire records with information on ignition and extinction times, administrative location (district, county, parish), burnt area, land cover type (forest and shrubland), and recorded cause of the fire.

Following the recommendations in Pereira et al. [1], we introduce a minimum burnt area threshold of `area_total` ≥ 0.1 ha to ensure temporal completeness and comparability of the data over the full 1980–2005 period. Smaller fires are known to be under-reported in the earlier years of the database, so restricting the analysis to fires with at least 0.1 ha of burnt area reduces artefacts caused by changes in detection and reporting practices. After applying this filter, we retain only records with non-missing values in the variables required for our subsequent modelling and XAI analyses.

In addition to reproducing the pre-processing choices documented in the original PRFD description, we identify an extra cause code 0 present in the working dataset but not discussed in Pereira et al. [1]. To maintain consistency with the published categorisation of fire causes, we explicitly re-label this value as an *Unknown* cause category before constructing the target variable.

1.2 Target variable

The predictive task focuses on a binary classification problem where the target distinguishes fires with an *intentional* cause from all remaining cases. Based on the cause codes provided in the PRFD, we group together all categories deemed to reflect deliberate human ignition into the positive class (`intentional` = 1), while accidental, natural and unknown causes form the negative class (`intentional` = 0). As mentioned above, the additional cause code 0, which is not documented in Pereira et al. [1], is treated as part of the *Unknown* group and therefore included in the non-intentional class. This definition produces an imbalanced binary target but reflects the operational distinction between intentional and non-intentional fires required for the subsequent XAI analysis.

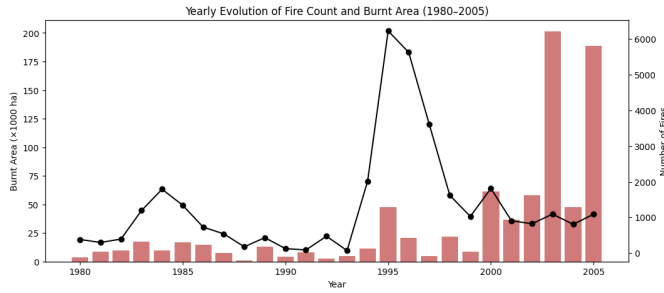


Fig. 1: Yearly evolution of fire count and total burnt area (1980–2005).

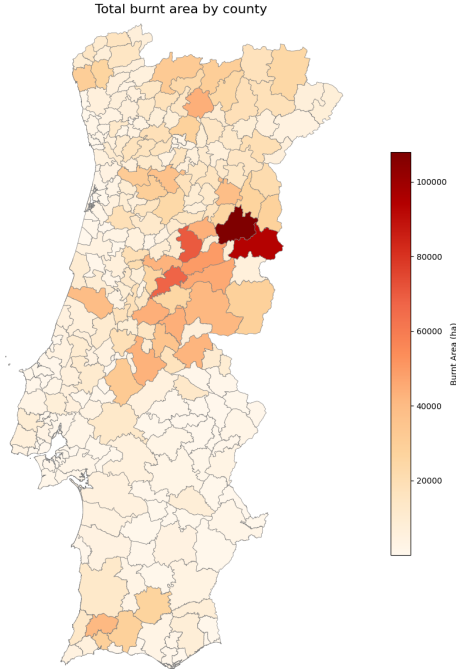


Fig. 2: Total burnt area by county in mainland Portugal (1980–2005).

1.3 Exploratory temporal and spatial patterns

Before fitting any predictive model, we examine the temporal and spatial structure of the PRFD subset used in this study. Figure 1 summarises the yearly evolution of the number of recorded fires and the corresponding total burnt area between 1980 and 2005. Both quantities display an overall increasing trend, with marked peaks in the mid-1990s and early 2000s, indicating periods of particularly severe fire seasons. This confirms that the analysis is carried out in a context of growing fire activity over time.

To characterise spatial heterogeneity, Figure 2 maps the cumulative burnt area by county across mainland Portugal. The largest impacts are concentrated in a small set of inland counties in the central and northern regions, whereas many coastal and southern counties exhibit comparatively low burnt areas. The map was obtained by joining PRFD county names

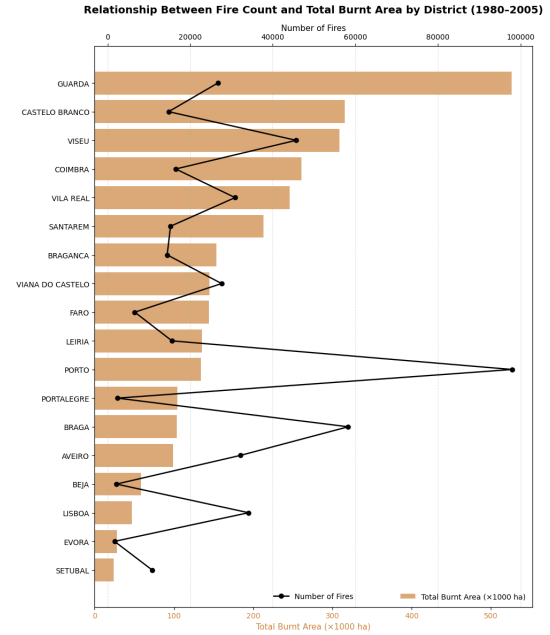


Fig. 3: Relationship between fire count and total burnt area by district (1980–2005).

to an official administrative shapefile; a small number of unmatched counties have burnt areas close to the national median and do not alter the overall pattern.

Figure 3 compares, for each district, the total burnt area (horizontal bars) with the number of recorded fires (superimposed line). Northern and coastal districts such as Porto and Braga register the highest fire counts but relatively modest burnt areas, suggesting frequent but mostly small events. In contrast, interior districts such as Guarda, Castelo Branco and Viseu show fewer ignitions but much larger total burnt areas, indicating rarer but more severe fires. This asymmetry between frequency and impact motivates the inclusion of both spatial and size-related variables in the subsequent modelling and XAI analyses.

When grouping fires into size classes, the vast majority of events fall below 10 ha, whereas a small fraction of very large fires accounts for most of the total burnt area (heavy-tailed size distribution).

1.4 Pre-modelling XAI techniques

Before fitting any predictive model, we applied unsupervised, pre-modelling XAI analyses to the numerical feature space. All continuous variables were imputed with column-wise means and standardised to zero mean and unit variance, so that differences in scale or missingness do not dominate the exploratory structure.

We first performed a Principal Component Analysis (PCA) on the pre-processed data. The cumulative explained variance curve (Figure 4) shows that a small number of components already captures most of the variability, indicating strong linear dependencies among the original descriptors. The leading

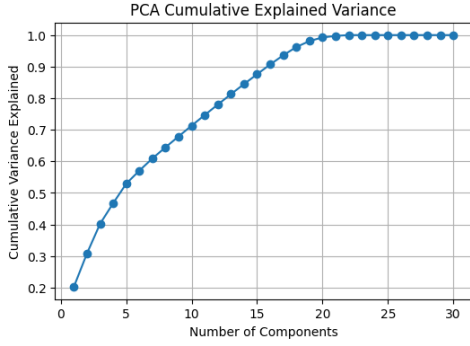


Fig. 4: Cumulative variance explained by the PCA components.

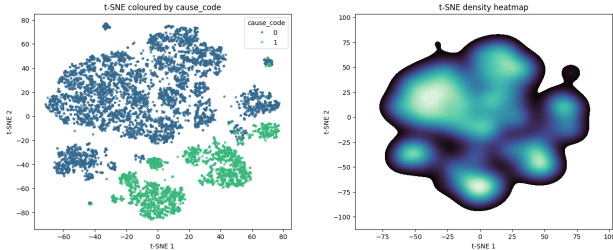


Fig. 5: t-SNE embedding of rural fire events, coloured by intentional vs. non-intentional ignition.

component is mainly associated with fire size and fuel availability, whereas subsequent components capture contextual variation not explained by burnt area alone. The projection onto the first two components reveals a dense central cloud with a few extreme points corresponding to very large events.

To explore non-linear organisation, we then computed a t-distributed Stochastic Neighbour Embedding (t-SNE) on a random subsample of the standardised observations, colouring the embedding by the binary intentionality label (Figure 5). The map exhibits several dense clusters, but intentional and non-intentional fires are largely intermixed, suggesting that the two classes are not trivially separable from the raw descriptors. Finally, we used K-means clustering on the same feature space to identify a small set of representative *prototypes* and rare *criticisms*, which will later serve as concrete reference events when analysing model predictions and post-hoc explanations.

TASK 2 – IN-MODELLING XAI

2.1 Geographic encoding and resampling

Before fitting glass-box classifiers, the cleaned dataset was further standardised: continuous attributes were imputed with column-wise medians and scaled to zero mean and unit variance. The target variable *intentional* is a binary indicator (1 = intentional ignition, 0 = all other recorded causes).

Spatial structure is encoded at district and county level only. Districts are represented by one-hot dummy variables. Counties are mapped to integer identifiers using a frequency-based scheme: frequent counties keep distinct codes, while infrequent ones are collapsed into a single baseline category.

This yields a compact geographic representation that preserves major spatial patterns without inflating dimensionality.

The training set is markedly imbalanced, with intentional fires representing approximately 16% of the observations. To mitigate bias towards the majority class, we apply SMOTE-based oversampling only on the training data: SMOTENC on the mixed numeric–categorical space (treating fire type and county encoding as categorical) and standard SMOTE when only numerical features are used. The test set remains untouched and imbalanced for realistic evaluation.

2.2 Glass model training

TABLE I: Performance of glass models.

Model	Geo	Acc	Prec ₁	Rec ₁	F1 ₁
LR	D+C	0.60	0.22	0.60	0.32
LR	C	0.57	0.22	0.66	0.33
DT	D+C	0.80	0.42	0.69	0.52
DT	C	0.75	0.37	0.78	0.50
RuleFit	D+C	0.84	0.52	0.32	0.39
RuleFit	C	0.85	0.62	0.11	0.19
BRL	D+C	0.65	0.31	0.95	0.46
BRL	C	0.63	0.30	0.96	0.45

Table I summarises the performance of all intrinsically interpretable classifiers. We consider three families of glass models: linear (LR), tree-based (DT) and rule-based (RuleFit, BRL) and, for each family, two alternative geographic representations: districts and counties jointly (D+C) and counties only (C). This design allows us to probe how much additional spatial resolution (district dummies on top of counties) effectively contributes to predictive performance and interpretability. All models were trained on oversampled data and evaluated on the original imbalanced test set.

Across methods, the expected recall–precision trade-off on the intentional class emerges clearly. The linear baseline (LR) remains the weakest option, with low precision and only moderate recall, reflecting limited capacity to capture non-linear interactions. Decision Trees substantially increase recall for intentional fires at the cost of more false positives, moving towards a high-sensitivity operating point. RuleFit behaves more conservatively: it achieves relatively high precision but very low recall, thus missing many intentional cases. Bayesian Rule Lists reach the highest recall, close to perfect sensitivity, but with low precision due to systematic overprediction of the minority class. Overall, DT and BRL are preferable when the priority is to detect as many intentional fires as possible, whereas RuleFit favours a stricter, low-recall regime. Differences between D+C and C configurations are modest: county-level information already captures most of the geographic signal, with district dummies providing only marginal gains.

In-model importance analyses display a consistent pattern across all glass models. Fire type is the dominant predictor of intentionality, followed by temporal descriptors (ignition and extinction dates, calendar attributes such as year and month) and the encoded county identifier. Vegetation and size variables (shrub, forest and total burnt area) provide additional signal, while district indicators and data-quality flags show low

global importance, appearing mainly as refinements in a subset of local rules. These in-modelling importance profiles define a transparent reference against which post-hoc explanations of the black-box model will later be compared.

TASK 3 – POST-MODELLING XAI

Post-hoc explanations were computed for a single reference black-box model. We compared three classifiers with both districts and counties: Random Forest, XGBoost, and a Deep Neural Network. The XGBoost classifier was selected as the reference model following a rigorous two-stage hyperparameter tuning workflow (Randomized Search followed by Grid Search). This tuning process was made to obtain a well-calibrated classifier, resulting in a substantial improvement in the F1-score (0.7780) for the minority class (intentional fires), which far surpasses any of the glass-box classifiers, and an overall accuracy of 86.80%.

The decision to use XGBoost over the Random Forest (accuracy 84.10%) and Neural Network (accuracy 82.58%) is justified by its superior balance between precision and recall for the critical class.

3.1 Simplification-based Global Surrogates

To obtain a global explanation of the opaque XGBoost ensemble, we train a global surrogate models. These are interpretable models trained to approximate the predictions of the black-box model rather than the ground truth. We implemented two distinct surrogates: a non-linear CART Decision Tree and a linear Logistic Regression. Fidelity is assessed by comparing surrogate and black-box outputs, the main metric is the agreement rate.

a) *CART Decision Tree vs. Linear Surrogate:* We trained a CART decision tree ($\text{max_depth} = 20$) and an ℓ_2 -regularised Logistic Regression on the pre-processed features, using the XGBoost outputs as targets. The fidelity comparison in Table II shows that the CART surrogate attains a clearly higher agreement rate and substantially lower LogLoss than the linear surrogate, indicating that the black-box model exploits non-linear structure that a linear decision boundary cannot reproduce.

The CART feature importances are dominated by continuous and temporal variables, namely *forest_total*, *ign_date_serial*, *fire_type* and *ext_date_serial*, with county and district indicators playing a secondary role. In contrast, the linear surrogate places its largest coefficients on *fire_type* and several district dummies (e.g. PORTO, BRAGA, COIMBRA, VISEU), complemented by data-quality flags and temporal variables, thus providing a simpler but less faithful global view of the same decision function.

TABLE II: Fidelity Comparison of Surrogate Models

Surrogate Model	Agreement Rate	Log Loss
CART Decision Tree	0.9232	0.1211
Logistic Regression	0.8369	0.3848

3.2 Feature-Based Explanations

Feature-based post-hoc analyses were computed using LIME, SHAP and PDP/ICE curves to characterise the behaviour of the tuned XGBoost model. LIME was applied to a representative test instance (index 3539), predicted as intentional with probability 0.88, and produced a sparse linear surrogate dominated by geographic dummy variables, together with *fire_type* and *ign_date_serial*. This reflects the tendency of local linear models to overemphasise binary spatial indicators within a small neighbourhood.

SHAP provides a complementary and more stable perspective. The global Shapley values, which were calculated over the test set, identify temporal and fuel-related variables particularly *ign_date_serial*, *forest_total*, *shrub_total* and *fire_type* as the most influential predictors across the dataset, while district indicators have consistently minor effects. For the same instance analysed with LIME, the SHAP waterfall plot highlights *ign_date_serial* and *forest_total* as the main contributors pushing the prediction towards the intentional class.

Sensitivity analysis via PDP and ICE confirms the limited marginal effect of *forest_total* and *shrub_total*, while revealing clear temporal structure. The probability of intentional ignition increases during summer months and shows a marked dip for events between 1987 and 1993, consistent with the historical introduction of specialised fire investigation brigades that improved cause attribution. Overall, SHAP and PDP/ICE indicate that temporal patterns dominate the global behaviour of the model, whereas LIME captures more local, and sometimes unstable, effects driven by sparse geographic encodings.

3.3 Example-based Explanations

We employed example-based methods to ground the model’s predictions in concrete data instances, utilizing a nearest-neighbour approach within the model’s internal feature space to ensure the plausibility of all generated explanations and respect domain constraints (e.g., mutually exclusive administrative regions).

The analysis of **prototypes and criticisms** (Table III) reveals structural instabilities hidden by probability scores. While the robust instance has zero critics, the borderline instance presents a conflict: despite a low intentionality probability (2.35%), it is surrounded by 3 critics from the opposite class. This explains the sensitivity to perturbations observed earlier. To validate the structural diversity of these explanations, we projected the learned prototypes onto the dataset’s principal components (Figure 6). The visualization confirms that prototypes are distributed across distinct dense regions, ensuring coverage of diverse fire typologies.

This sensitivity is quantified through **Counterfactual Explanations**. As shown in Table IV, for a high-confidence Intentional instance (98%), the counterfactual search requires changing immutable features to flip the prediction to Non-Intentional. Specifically, a shift from a weekday (Thursday) to a weekend (Sunday), combined with a change in location

TABLE III: Prototype and Critic based analysis.

Instance (Arch.)	Prob. (1)	P/C	Local Insight
Robust Non-Intentional	0.02%	3/0	Stable: Deep in safe region. No opposing critics found.
Sharp Intentional	98.24%	3/2	High Conf.: Strong signal, but 2 critics imply boundary proximity.
Borderline	2.35%	3/3	Unstable: Low probability but max critics (3). High risk of flipping if perturbed.

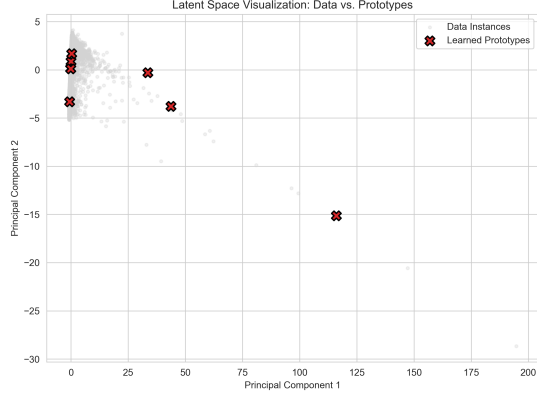


Fig. 6: Latent space visualization (PCA) of the dataset and learned prototypes. The prototypes (marked in red) effectively cover distinct regions of the data manifold, supporting the retrieval of diverse examples for local explanations.

(Viana do Castelo to Viseu), is required. This dependency on the *District* variable highlights a potential geographic bias, as the model relies on immutable context rather than solely on actionable features like vegetation management.

TABLE IV: Counterfactual explanation analysis: flipping the prediction from Intentional (98%) to Non-Intentional.

Feature	Original	Counterfactual	Type	Insight
Ignition Hour	12:00	13:00	Actionable	+1 Hour shift
Weekday	Thursday (3)	Sunday (6)	Actionable	Mid-week to Weekend
District	Viana do Castelo	Viseu	Immutable	Geo Bias Revealed
Fire Type	Reignition	Reignition	-	No Change

Finally, the **Anchors algorithm** highlighted a complexity gap between classes (see Table V). The model identifies non-intentional fires using simple heuristics, such as the rule $\text{fire_type} > 0 \wedge \text{district_PORTO} > 0$ (95% precision). Establishing intentionality, however, requires satisfying a complex conjunction of conditions involving *forest_total*, *shrub_total*, and seasonality. This demonstrates that the model views intentionality not as a default state, but as a specific exception occurring only when precise environmental and temporal factors align.

TASK 4 – QUALITY OF THE EXPLANATIONS

To assess the reliability and utility of the generated explanations, we adopted a functionally-grounded evaluation

TABLE V: Anchor rules extracted for Intentional and Non-Intentional fires.

Target Class	Anchor Rule	Prec.	Cov.	Insight
Non-Intentional (0)	$\text{fire_type} > 0 \wedge \text{district_PORTO} > 0$	0.95	0.11	Simple Pattern: Reignition in Porto is almost certainly accidental.
Intentional (1)	$\text{fire_type} \leq 0 \wedge \text{forest} > 0 \wedge \dots$ (6 conditions)	0.97	< 0.05	Complex Pattern: Intentional fires require specific environmental & temporal contexts.
Intentional (Relaxed)	$\text{fire_type} \leq 0 \wedge \text{forest_total} > 0$	0.86	-	Core Driver: First ignitions in forest areas are the primary risk factor.

approach. This methodology utilizes objective, proxy-based metrics to quantify properties such as faithfulness, sparsity, and stability, avoiding the subjectivity inherent in human surveys. A specific methodological constraint was applied regarding the sample size. While SHAP, LIME, and Surrogate evaluations were computed over a representative sample of 1000 test instances to ensure statistical robustness, the evaluation of the Anchors algorithm was restricted to a subsample of 100 instances. This decision was necessitated by the extreme computational cost of the method, which averaged approximately 3.9 seconds per instance, rendering a large-scale evaluation infeasible within the project constraints compared to the sub-millisecond execution of the other methods.

The quantitative results, summarized in Table VI, highlight a distinct trade-off between fidelity, interpretability, and computational efficiency. Regarding faithfulness, which measures how accurately the explanation reflects the underlying model, SHAP demonstrated the highest performance with a rank correlation of 0.51. This indicates that the feature importance rankings produced by SHAP align closely with the model’s actual sensitivity to feature removal. However, this fidelity comes at the cost of complexity. SHAP utilizes an average of 42.0 features per explanation, as indicated by the sparsity metric. Presenting a user with over forty variables creates a dense information environment that imposes a heavy cognitive load, potentially obscuring the most relevant drivers of the decision.

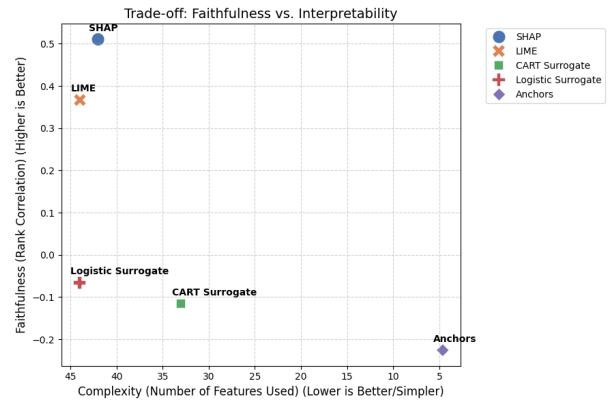


Fig. 7: Trade-off analysis between Faithfulness (Rank Correlation) and Complexity (Sparsity). SHAP achieves high fidelity but requires complex explanations, whereas Anchors offer maximum simplicity at the cost of lower fidelity.

TABLE VI: Functionally-Grounded Evaluation Metrics for XAI Methods

Method	Runtime (s)	Fid. MSE	Fid. R^2	Rank Corr	Sparsity	Entropy	Cov.	Stability	Lipschitz	Consist.	Comp.	Suff.	Plaus.
SHAP	0.000762	—	—	0.510242	42.00	2.712036	—	2.553463	0.066775	1.000000	-0.650255	0.162619	1.96e-08
LIME	0.028004	—	—	0.366759	43.97	3.394562	—	0.951232	0.021949	0.541715	-0.244588	0.476765	1.96e-08
CART Surrogate	0.000018	0.037705	0.704154	-0.115787	33.00	2.462545	0.278356	0.000000	0.000000	-0.166707	-0.416140	0.306439	1.96e-08
Logistic Surr.	0.000955	0.055663	0.563251	-0.065909	44.00	2.806278	—	0.000000	0.000000	0.184165	-0.464803	0.258130	1.96e-08
Anchors	3.913403	—	—	-0.225666	4.64	1.351525	—	2.706981	0.107835	—	-0.439155	0.218618	1.96e-07

In contrast, the Anchors algorithm proved to be the most interpretable method from a cognitive perspective, utilizing only 4.64 features on average and achieving the lowest entropy score of 1.35. However, Anchors exhibited a negative rank correlation of -0.23. We argue that this metric is ill-suited for logic-based methods, as it penalizes the binary nature of rules (features are either present or absent) against the continuous feature attribution rankings assumed by Spearman correlation. Therefore, the validity of Anchors should be assessed via its high Precision VI rather than Rank Correlation. Nevertheless, the runtime analysis reveals a prohibitive disparity. The latency of the Anchors algorithm renders standard implementations unsuitable for real-time decision support in high-throughput environments. Furthermore, regarding robustness, global surrogates offered perfect stability, whereas local methods like Anchors and SHAP showed significant sensitivity to input perturbations, identifying a potential trust issue in noisy environments.

Based on these functionally-grounded metrics, we suggest two specific strategies to optimize the interpretability pipeline. To address the cognitive overload caused by the high sparsity of feature-based methods, we recommend enforcing a visualization filter that displays only the top-5 features for SHAP. This approach would trade a negligible amount of completeness for a substantial gain in user readability. Additionally, for operational scenarios where computational speed and stability are paramount, replacing the computationally expensive Anchors method with a pruned global CART surrogate limited to a shallow depth offers a viable compromise. As evidenced by the evaluation metrics, surrogates provide instant execution and perfect stability, offering a balanced alternative between rule-based simplicity and operational requirements.

CONCLUSIONS AND FUTURE WORK

This work goes beyond the sole objective of predicting the fire intentionality and instead uses this task to demonstrate the importance of using a XAI methods in complex, real-world datasets, where the decisions boundaries are not well behaved and can't be captured by simple understandable models. The performance of the glass-box classifiers, particularly their low F1-score on the intentional class, indicates that intentional and non-intentional fires are not trivially separable. This confirms the need for expressive black-box models when addressing fire cause prediction, despite the loss of intrinsic interpretability.

The analysis of both glass-box and black-box models confirms that the key drivers of fire intentionality are consistent across complexity levels, dominated by fire type, total forest area, and temporal features (especially ignition date and year).

However, the interpretability methods reveal distinct trade-offs: while the Linear Surrogate offers maximum simplicity by collapsing the logic primarily onto *fire_type* and specific districts, it sacrifices fidelity by ignoring crucial non-linear interactions. In contrast, the high-fidelity methods like SHAP and XGBoost's intrinsic importance profiles consistently validate the dominance of forest fuel and timing variables. Local explanations like LIME and local SHAP further confirm that the model's complex decisions for a specific fire event rely on instance-specific combinations of these few highly influential features. This unified view reinforces the need for a multifaceted XAI approach to truly understand the predictive decision process.

The proposed pipeline further highlights the necessity of systematically evaluating the applied XAI techniques. None of the considered explanation methods dominates across all metrics. For instance, SHAP achieves the highest faithfulness, making it a strong candidate for accurately reflect the underlying model. However, its high sparsity limits its usability for non-expert users, as its explanations involve a large number of features. In contrast Anchors highly sparse and thus easily interpretable explanations, but their high computational cost makes them impractical to explain the model globally. These results underline that XAI methods should be viewed as complementary tools, its own unique use-case rather than as universal optimal solutions

Finally, this work demonstrates the value of functionally-grounded evaluation metrics for comparing explanations methods in an objective manner, without relying on human feedback. While such metrics enable precise and efficient assessment, these explanations ultimately aim to help humans understand the black box-models decisions. As a result, an important direction for future work is to investigate the alignment between functionally-grounded metrics and human-centered evaluations, in order to determine whether these quantitative measures reliably capture the perceived quality and usefulness of explanations in real decision-making contexts.

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